Project Misfits Design Bible

This document centralizes documentation for design decisions made in each domain. It mainly documents gameplay design decisions and UI/UX design decisions, with other domains linking to their respective bibles. If you have any questions, comments, or concerns, reach out to Jeremy King.

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Player Ability Design

TODO for this document

1. Document non-hazardous interactable design (e.g. door keys, levers, and the like)

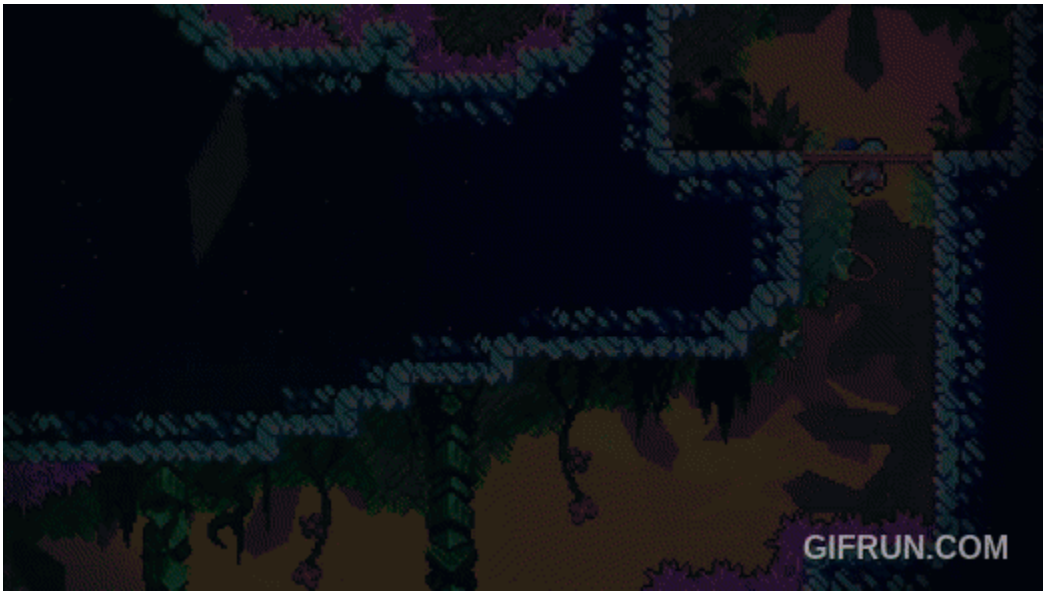
Design documentation

Overall, Player movement will feel responsive, floaty, and fluid. The Leaf Dash should feel fast, evasive, and highly versatile but restricted by resource management; the need to manage the Leaf Meter.

- Run
  - Moves Fenn horizontally.
  - Reaching max speed is quick and snappy. High acceleration, deceleration & turn speed.
  - Max speed is decently fast. Slower than Mario, closer to Hollow Knight or Celeste.
- Jump
  - Moves Fenn vertically.
  - “Floaty” jump, Fenn stays airborne for a long time relative to other games.
  - High mobility in the air: turning is quick, allowing the player to easily correct when jumping too far or too little.
  - Jump input can be released to “fast fall”, multiplying gravity on the player to, well, fall faster.
    - This allows the Player to have variable jump heights by releasing the jump button at different points in the jump.
- Leaf Meter
  - While moving, Fenn continuously builds up “wind” in a meter. Fenn does NOT build up “wind” while they are turning.
    - Similar to the P-Meter in some Mario games

  - While NOT moving, the Leaf Meter slowly depletes over time.
    - This pushes players to stay moving constantly to keep their Leaf Meter high and prevents the exploit of building up the Leaf Meter in a safe area then never having to worry about building up the meter again.
  - The Leaf Meter continuously depletes at a faster rate when Fenn is Leaf Dashing.
  - As such, the Leaf Meter represents the player’s momentum and overall time in motion, rewarding the player with more Leaf Meter when they are at top speed for longer periods of time.
- Leaf Mode
  - Fenn transforms into leaves, allowing them to get through small gaps or dash through the air depending on the context.

- Leaf Pile: If no direction is held when activated Leaf Mode or the player has no Leaf Meter, Fenn will turn into leaves and slowly fall to the ground. The player can slowly move, or “shimmy”, left or right.
- Leaf Dash: If a direction is held when entering Leaf Mode and the player has some amount of Leaf Meter, Fenn will turn into leaves and launch in the held direction at a very fast speed.
    - The player remains at a consistent high speed during the Leaf Dash and may move in a 360-degree radius.
    - Turning during the Leaf Dash has a turn radius that means players must be fluid with their movement.
    - When the Leaf Meter is nearly empty, Fenn’s velocity and turn speed will begin decreasing.
    - Requires Leaf Meter to use and will last for as long as the input is held OR until the Leaf Meter is fully depleted.
        - The Leaf Meter does NOT need to be full to Leaf Dash. It only needs to be partially filled (>0%).
        - The Leaf Meter continuously depletes at a faster rate when Fenn is Leaf Dashing.
    - Fenn is invulnerable while Leaf Dashing and cannot take damage from enemy attacks.
    - The Leaf Dash enables the player to launch Fenn across gaps, up vertical platforming, through small tunnels, and more.



- Command Companion
    - Tell Fenn’s companion to perform a contextual action. This may be moving to the nearest interactable object, following Fenn, distracting an enemy, etc.
    - The action provides some benefit to Fenn and is essential to completing the room, area, or level.

- Interact
    - Contextual action which “interacts” with a nearby object or NPC.
    - If “interacting” with an NPC, triggers a conversation between the two characters.
    - May also “interact” with levers, keys, and other objects.

BIG DESIGN THINGS FOR PLAYER:

- we need to make the player not stop super hard after landing
- Make player jump short but long
- fix the small speed up right before you land
- Make fans that push the player while in leaf mode
- Open areas for the leaf dash
-
- 1 tile = 10 units
- 650 units = landing distance for fling (obstacles should be placed a little
- 16 units wide = safe jump max
- 80 units high = good jump heights
- 70 units = good gap for post jump ceiling spikes

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Obstacles and Threats Design

TODO for this document

1. Document the design of enemies and obstacles that will appear throughout the level
2. Document the design of small and challenging platforming patterns
3. Document the design of player & companion abilities and how they address those challenges

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Design documentation:

Overall, the obstacles and threats the player will face with Fenn pose a threat to one of three things: platforming progress, Fenn's health, or Fenn's Leaf Meter.

Goals for Enemies:

- Should act more like obstacles
- Player should not be expected to directly fight enemies

Goals for Companion:

- Companion should allow the player to 'combo' with whatever they are doing to effect the environment and obstacles
- Should be made clear the companion is only able to affect things due to who they are (i.e Nothing that's just flipping levers).

Goals for Environmental Obstacles:

- Should feel very rhythmic
- Should reward players for taking more dangerous paths
- Should be dynamic/not repetitive

Platforming obstacles and threats:

- Slippery platforms
- Large gaps
- Falling icicles
- Moving platforms

Enemy obstacles and threats:

- Rolling snowballs
- Sliding ice cubes
- Little flying mosquito guys that shoot icicles (like those squits or whatever from Hollow Knight)
- Spike balls
- Ice Goose

Companion Abilities:
Bangles of Reach:

- Az has a pair of magical bangles that she wears with that can as portals to one another. This means that while she still has a bangle the other can act like an extension of her arm.

- She gives Fenn one of her bangles allowing him to summon of her arms to hold something in place
- Things that the hand can affect will have a red claw mark
- For example, holding a charging enemy in place, which rushes forward when the claw is released
- For example, holding a swinging platform in place to allow the player to use it as a platform

Environmental obstacles or tools that actively interact with the player

- Platforms that drain wind meter while standing on them

- Enemies or objects that contain a bunch of air that when broken open somehow releases that air in a specific direction that launches the player

- Fan objects that push the player in a specific direction, potentially increasing the wind meter while the player is in its path

- Enemies or objects that contain a bunch of air that when broken open somehow releases that air in a specific direction. While within the gale that is released the player has infinite wind meter

- Mail pipes that when interacted with in leaf mode will send the player to another location

- Enemies that only track and can touch the player when they are not in leaf mode and enemies that are the exact opposite. When these two enemies collide they explode

- Icicles that fall from the ceiling on an interval that can be held in place by the bangle ability

- Spikes that hurt the player both in leaf mode and non leafmode

- Ice Cube charges at the player, they will stop charging once they get to where the player was. This is unless you use the claw in which case they will push extra hard and then zoom until they hit a wall.

Themes for the Three Heaters:

- Ice Cube enemies
- Wind Puffs
- Wind Tunnels

Planned Obstacles and Threats

Obstacle	**Fenn's ability to address it** (Should only have basic platforming, Leaf Pile/Leaf Dash, and Claw ability actions)	**Result**
Wide horizontal pit/gap with a hazard	Leaf Dash	Fenn maneuvers over the gap and avoids the hazard

Block that falls when you stand on it	Claw ability is highlighting moving block -> trigger claw ability	Block freezes in place until claw ability is released or Fenn is no longer in range. Fenn can continue to stand on the block
Block that falls when you're underneath it	Claw ability is highlighting moving block -> trigger claw ability	Block freezes in place until claw ability is released or Fenn is no longer in range. Fenn can stand on the block if it's reachable
Leaf Dash-only mail pipe	Leaf Pile Leaf Dash	When in Leaf Pile, Fenn can crawl into pipe and slide along to its output (think Super Mario 3D World glass pipes) When in Leaf Dash, Fenn can direct their momentum into the pipe and slide along to its output
Slippery platforms	Platforming Leaf Dash	Fenn slides around when in contact with platform Fenn uses the Leaf Dash to skip using the platform
Wind draining platforms	Platforming Leaf Dash	Fenn's wind is drained while in contact with platform Fenn uses the Leaf Dash to skip using the platform
Charging ice cube enemy	Claw ability is highlighting charging enemy -> trigger Claw ability Jump over enemy Leaf Dash	Charging enemy freezes in place until claw ability is released or Fenn is no longer in range Fenn platforms around the enemy Fenn uses the Leaf Dash to avoid being hit by the charging enemy
Floating hunter enemies (swap between chasing the player and inert based on if the player is in Leaf Mode or not)	Leaf Mode Platforming	Entering/exiting Leaf Mode switches the hunters' state. Any that were chasing the player become inert and any that were inert start chasing the player Fenn outmaneuvers the enemies

	Leaf Dash	with platforming Fenn outdistances the enemies with the Leaf Dash

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Level and Room Design

TODO for this document

1. Document room design decisions
    - What will the player see when entering a room?

Design documentation

Level Types

- All levels are areas of the kingdom's castle, the Bay Leaf Palace.
- The first level contains the Council Chamber and the Great Hall. The Council Chamber is where the knights are sealed, so the player cannot actually enter it. The Great Hall is the area where the player first gets control of Fenn, where can speak to NPCs about what's going on, where the player meets Az, where the entrances to gameplay-oriented levels are, and so on. For our purposes, this level is simply referred to as the Great Hall.
- The first gameplay level is called the Cellar.

Room Types

- Zero challenge rooms
    - Connecting hallways/areas
    - Respite rooms (like benches in Hollow Knight)
- Low challenge rooms
    - Minimal platforming and/or enemies that pose a small threat to the player
- Medium challenge rooms
    - The player must use their agility and abilities to overcome platforming and/or enemies that pose a direct threat to the player
- High challenge rooms
    - The player must be comfortable with the various abilities at their disposal to overcome a major platforming, enemy, or other threat
- Special challenge rooms
    - The player must use their abilities to overcome a challenge that furthers the story/quest
    - Difficulty is rated Medium to High
    - For example, using their platforming abilities to scale slippery platforms and pull a lever, turning on a heater
- Boss room
    - The player must use all of their abilities and all of the knowledge gained facing challenges throughout the level to escape a final test of skill
    - Also a challenge that furthers the story/quest (naturally)

Example rooms:

- A long room where the player must build up their Leaf Meter and use it to cross a wide pit of spikes on the floor
- A tall room where the player must build up their Leaf Meter and use it to maneuver vertically and reach a high ledge

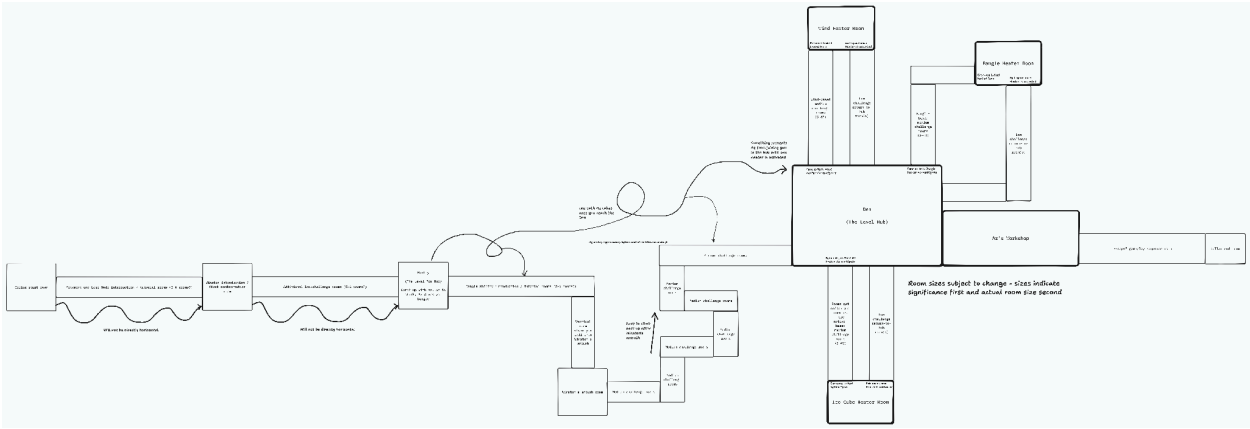
- A closed-in, donut-shaped room where an enemy moves around in a circle and the player has to navigate around the donut shape to avoid it

Level Structure, Layout, and other Details

- The structure and layout of the level strongly correlates with the level's narrative structure, which is described in more detail in the Narrative Outline here**.** A level design-oriented overview is also provided below.
- There will be no branching paths in the level. The only time the level will branch is when the player reaches the level hub and must enable each of the three heaters.
- Rooms will be navigated between via doors the player presses a button to interact with, as opposed to screen transitions you see in games like Hollow Knight.
- Rooms will only ever have one entrance and one exit. Specific level layout related rooms (like the GreatHall Hub and the Cellar Hub) will have more than two doors. Speak with Jeremy King if you are working with a room that needs more than two doors.
- There will be no hidden/destructible walls in the level. All traversable areas should not be obscured. That isn't to say that secrets can't be included; they just need to be in corners of existing rooms and make use of the game's existing obstacles and threats rather than requiring the player Leaf Dash into a random wall that's actually fake.
- Rooms should gradually increase in challenge. The first rooms will be zero challenge rooms to get the player accustomed to the movement and abilities, with the latest rooms primarily being medium and high challenge rooms.
- The level will follow kishotenketsu, where mechanics are presented to the player in the following stages: introduce, develop, twist on concept, and master.
    - The rooms between the level hub and the heater rooms will function as the twist on concept challenges and the heater rooms will function as master challenges.
    - Mechanics can follow the kishotenketsu method at different paces as needed. For example, enemies will not be introduced to the player immediately. They will show up once the player has completed a couple rooms that introduce and develop the movement mechanics of platforming and Leaf Mode. Same goes for the Claw companion ability, which will be introduced and developed roughly halfway between the start of the level and reaching the level hub room.
- Heater rooms and the boss room will be built out of existing obstacles and threats rather than custom designed/coded sequences.
    - Building the special challenge and boss challenge rooms out of our basic obstacles and threats simplifies the development and design process.

Jacobs Level Design Findings:

Level Overview:



Because the image is large, it is also uploaded to the Drive. You can zoom in and actually read the text and other details here:
LevelLayout.png

Room Pipeline

Room files (each being an individual Godot scene) will use the following naming convention:
01_LevelName_a_RoomName_room.tscn

- '01' refers to the internal level number. GreatHall is 01 and Cellar is 02.
- 'LevelName' refers to the name of the level itself. This will be one of the names listed above in the Level Types section. This section must use CamelCase for legibility.
- 'a' refers to the portion of the level that the room appears in. The starting tutorial might be 'a', the companion ability tutorial section might be 'b', the heater portions might be 'c', 'd', and 'e'. These are to help us see the general progression of the level with just a glance at the filenames.
- 'RoomName' refers to the name of the room itself. 'Entrance', 'Boss', and 'IciclePlatforming' are all acceptable names. This section must use CamelCase for legibility. Try to stick to one-word names, but it's fine if more than one word is needed to describe the room at a high level.
- All rooms must have the suffix '_room.tscn'. This makes it easier for us to filter out what scene files are rooms. The door entity has a variable that points to what room they go to, which uses the file path to the room scene file itself. Ending the files with '_room.tscn' means it will show up in the file picker when changing this parameter in the inspector tab for each door.
- Grayboxed levels instead must follow the format 'GB_RoomName_room.tscn' and go in the folder 'res://src/rooms/graybox/'.
- Testing levels instead must follow the format 'TEST_RoomName_room.tscn' and go in the folder 'res://src/rooms/test/'.
- For some room name examples, see below:
    - 01_GreatHall_a_Entrance_room.tscn
    - 01_GreatHall_b_CellarDoor_room.tscn
    - 02_Cellar_a_AzEncounter_room.tscn
    - 02_Cellar_c_AzWorkshop_room.tscn

Creating a new room:

- In the project files, there is a scene at the path ‘res://src/rooms/room.tscn’. Right-click it in the FileSystem panel and select ‘New Inherited Scene’. It will open a new scene in the main editor and scene panel.
- Before making any changes, save the scene, and put the file in the appropriate level folder with the appropriate filename as outlined above.
- Next, double-click the ‘Room’ node in the Scene panel and rename it. Name it exactly the same way you named the file, but without the ‘_room.tscn’ at the end.

Setting up room transitions:

- Each Room scene has a child node called ‘Doors’ by default. This node will have two child Door nodes, one for the room entrance and one for the room exit. If you ever need to add a Door to your room (say you accidentally delete one of the existing ones), click the ‘Instantiate Child Scene’ button at the top of the Scene panel and select ‘door.tscn’. Make sure that every door you add is a direct child of ‘Doors’!
- Every door you add will need three variables set in its inspector panel before it works correctly. Each inherited Room scene starts with two doors to make this part easier.
    - Path to Target Room: This is the file path of the Room scene you want the door to lead to. Click the file icon on the right of the box and navigate to ‘res://src/rooms/<your level>/’. Select the file for the room you want the door to go to.
    - Door Name: This is the name of this door. This should be ‘enter’ or ‘exit’. Doors at the logical “start” of the room are ‘enter’ and doors at the logical “end” of the room are ‘exit’. Name the Door nodes in the scene tree ‘DoorEnter’ and ‘DoorExit’ to match for convenience.
    - Target Door Name: This is the name of the door you want the player to spawn at in the target room. If this door is ‘enter’, this variable should be ‘exit’. If this door is ‘exit’, this variable should be ‘enter’.

The Tech Bible describes “glue code” that will string together all of our game systems. The SceneManager is one such system, as it handles swapping rooms in and out, switching from menus to gameplay, and the like. It will not handle anything beyond that; any initialization those scenes need to do must be conducted by themselves.

Current Levels: 14
Narrative Rooms: 5

1. **Great Hall**

- **Tutorial room 1 (movement)**
- **Tutorial room 2 (leaf dashing 1)**
    - Experience goal: Introduce the mechanic and teach player to curve through places
- **Tutorial room 3(Leaf dashing tutorial 2)**
    - Experience goal: Introduce spikes and imply fling
- **Longer form room with only only running and dashing**
    - **Experience goal: Get player used to running and jumping and dashing with minimal amounts of danger**

2. **Meet Winston**

- **Windzone Tutorial 1**
- **Windzone tutorial 2**
    - **Experience goal: introduce player to bouncing between dash zones**
- **Longer windzone level 1**
    - **Experience goal: really get some fun less challenging douncing between dash zones**
- Longer windzone level 2:
    - Experience goal: more of a focus on going longer distances with the dash
- Longer Windzone Level 3:
    - Combine dashing between zones and longer distances within dash zone

3. **Pantry**

- **Intro to leaf platforms**
- **Leaf platforms level 1:**
    - **Experience goal: Combine with just normal and jumping**
- Leaf platforms level 2:
    - Experience goal: Combine with windzone
- Leaf platforms level 3
    - Experience goal: Combine with breakable platforms
- Leaf platforms level 4:
    - Experience goal: Combine with everything so far
- **Breakable platforms tutorial**
- **Breakable platform longer level 1**
    - **Experience goal: Combine with rooms that only require running and dashing to add some more pressure**
- Breakable platform longer level 2
    - Experience goal: combine with windzones distance things
- Breakable platform 3
    - Experience goal: combine with wind zones and bouncing
- Breakable platform 4
    - Experience goal: Mastery check

4. **Ambush (and then post-ambush)**
- **Ambush mini room**
    - **Experience goal: Teach players about the bangle but as like a one time thing not a full tutorial (COULD WE MAKE THIS A CUTSCENE?!?!?!?!**
- Bangle tutorial room
- **Bangle longer form room 1:**
    - **Experience goal: Emphasis on bangle opening paths for the player like up and down doors**
- Bangle longer form room 2:
    - Experience goal: Rotating platforms
- Bangle longer form room 3:
    - Experience goal: Combine with windzones
- Bangle level 4
    - Experience goal: Combine with breakable platforms

5. **Den**
6. Maze room
- **Pipe tutorial 1**
- Pipe level 2
    - Experience goal: Leverage pipes and the flinging for cool levels and to help get to different sections
- Pipe level 3
    - Experience goal: Combine with breakable platforms and windzones
7. Small room ("fireside chat 1")
- Intro to ice scream enemy
- Ice scream level 1:
    - Use as pressure tool
- Ice scream level 2:
    - Show they can kill each other (Loopy dy loops
- Ice scream level 3:
    - Combine with windzones
- Ice scream level 4:
    - Big mastery room
8. Heater
9. Small room ("fireside chat 2")
   5 smaller and easier levels that still combine a multitude of features

10. Den again (asthma attack)
11. Workshop
12. Boss Challenge --> Heater

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Narrative and Dialogue Design

Design documentation

Information about narrative design–such as narrative overview, character profiles, conventions, and more–can be found in the Narrative Bible.

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UI/UX and Interface Design

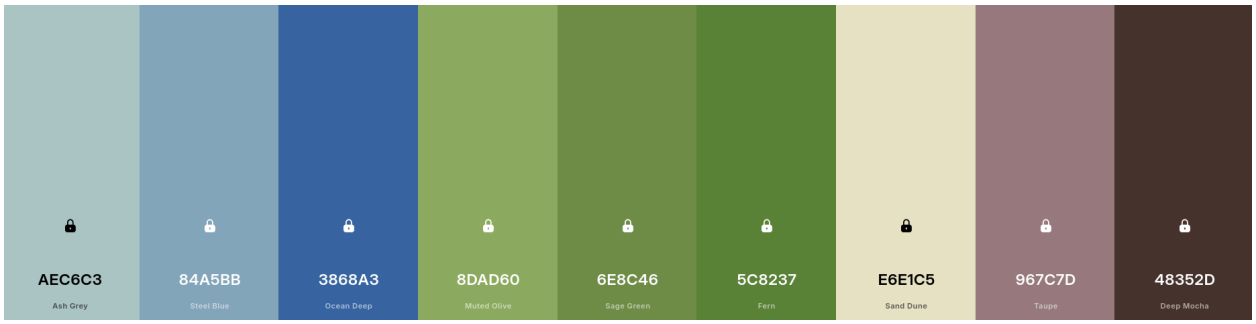
TODO for this document

1. Document UI standards across all assets and elements
    - Fonts
    - Sizing
    - Color Palette
    - General Style guidelines
2. Document what UI elements the game should have and how critical each piece is
3. Document UI/UX pipeline

Design documentation

Standards

- Fonts
    - Beholden (https://amorphous.itch.io/beholden?download)
        - Used for all text
        - All capitals for titles
    - ZT Mota (https://www.1001fonts.com/zt-mota-font.html)
        - Used for the game Logo and main menu title
- Initial Color Palette



- Specific Colors
    - UI Base Color: #E6E1C5
    - UI Base Text Color: #352620
    - Leaf Color (unless otherwise indicated): #5C8237
    - Title/Main Screen
        - Game Title Color: #4F8F0E
        - Return Address + Stamp Base Color: #AEC6C3
        - Return Address + Stamp Accent Color: #617875
    - Menu/Settings/Control Screens
        - Envelope Color: #BCB692
        - Title Text Color: #576F35
    - Dialogue Boxes
        - Fenn Name + Accent Color: #576f35

        - Az Name + Accent Color: #A86A19
        - Winston Name + Accent Color: #35639C
        - Wizard of Doors + Accent Color: #6B2D76
        - Plain Color: #352620
        - Progress Dialogue Indicator: #352620
    - HUD Health
        - Background Color: #967C7D at 65% transparency
            - NOTE: This may change once things get into engine because of it being overlaid onto the game but I wanted to have something that I could try to start.
    - HUD Wind Meter
        - Background Color When Not Full: Background Color: #967C7D at 65% transparency
            - NOTE: This may change once things get into engine because of it being overlaid onto the game but I wanted to have something that I could try to start.
        - Background Color When Full: #8DAD60 at 65% transparency
        - Swirl + Leaf Color: #425422

Critical Elements

- Title/main menu screen
    - Envelope
        - Stamp Takes you to credits
            - Cute icon in the middle with the “Credits” curved around on the top and the bottom
        - Return Address takes you to how to play
            - “KINGDOM’s How to Play”
        - Project title and subsequent selections underneath are centered
- Settings screen/ Controls screen
    - Will have letter theming
    - Will be an overlay that covers game play
        - Game play in background will be paused and blurred and muted in color
- Dialogue box
    - Inspired by Moorish architecture and shapes
    - Dialogue loads left to right
    - Blinking indication ability to continue on
- Player HUD (health, wind meter primarily)
    - Health
        - Discrete lives displayed by leaves that disappear to indicate life/health lost
    - Wind Meter
        - Progress bar

        - Long wind symbol will slide in from left to right with leaves that appear to show progress

Pipeline

- Art Creation
    - Base assets (no text) are created and exported as PNGs to be added in engine
    - All assets are created in a way that each piece can be extracted from each other
        - Ex: envelope base and stamp are created separately from each other and are exported as two assets
- Engine Implementation
    - Assets are imported into the UI assets folder and into subsequent folders based on use
    - All assets are placed together in their respective Godot scene allowing for reuse of assets
        - Background is placed into scene then button art is placed on top of the background
    - Any text is added last using Godot's built in text and button features.
    - Once all assets are placed, code and functionality can be added to the scene and each element when needed

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Art and Aesthetic Design

TODO for this document

4. Document visual design decisions for characters
5. Document visual design decisions for environments
6. Document design decisions for art style
7. Document design decisions for color palettes, textures, and other techniques and aesthetic choices

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Design documentation

Information about art design–such as visual style, brushes, conventions, and more–can be found in the Art Bible.

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Audio and Music Direction

TODO for this document

8. Documenting decisions about the soundscape and mix
9. Documenting music pipeline and SFX pipeline
10. Documenting Audio and Music decisions
11. Document audio conventions and tools
    - Music and SFX asset creation
    - Music and SFX systems and programming

Design documentation

Information about sound design–such as musical style, soundscape mood, conventions, and more–can be found in the Audio Bible.